



REPUBLIC OF THE PHILIPPINES
MUNICIPALITY OF TANAUAN, LEYTE

Public Market Rental Monitoring System

Market Office — Municipal Government

 **USER MANUAL**

Version 1.0 · August 2026

A comprehensive guide to operating every feature of the system

≡: Table of Contents

PART I — GETTING STARTED

1 Introduction & System Overview

1.1 About the System

1.2 Key Features at a Glance

1.3 System Requirements

2 System Interface & Navigation

2.1 Application Layout

2.2 Sidebar Navigation

2.3 Top Bar & Search

2.4 Notifications Panel

PART II — CORE MODULES

3 Dashboard

4 Stall Management

4.1 Viewing & Filtering Stalls

4.2 Adding a New Stall

4.3 Viewing Stall Details

4.4 Editing a Stall

4.5 Deleting a Stall

5 Tenant Records

5.1 Managing Tenants

5.2 Monthly Rent Collection

5.3 Early-Payment Discount (20%)

5.4 Stallkeeper Management

6 Applicant Management

6.1 Application Pipeline

6.2 Documentary Requirements

6.3 Approval & Stall Assignment

7 Utility Billing

7.1 Bill Calculator

7.2 Billing Records & Management

7.3 Receipt Printing & Batch Printing

7.4 Late Surcharges

8 Violations Register

PART III — REPORTING & INTELLIGENCE

9 Analytics & Operations Report

10 Logbook (Audit Trail)

11 Market Assistant (AI)

PART IV — ADMINISTRATION

12 Officers & Organizational Board

13 Stall / Space Verification Slips

14 Settings & Data Management

15 Support & Help Center

APPENDICES

A Business Rules Quick Reference

B Troubleshooting & FAQ

C Contact Information



CHAPTER 1

Introduction & System Overview

1.1 About the System

The **Public Market Rental Monitoring System (PMRMS)** is a desktop application developed for the **Market Office of the Municipality of Tanauan, Leyte**. It serves as the centralized digital platform for managing all operational aspects of the Tanauan Public Market, including stall management, tenant records, rent collection, utility billing, applicant processing, violation tracking, and official report generation.

100% Offline Operation: This system runs entirely on your local computer. No internet connection is required. All data is stored securely on your workstation and never transmitted to external servers, ensuring complete data privacy.

1.2 Key Features at a Glance

 Dashboard Real-time overview of market operations, revenue, occupancy rates, and recent activity.	 Stall Management Complete registry of all market stalls with status tracking, inspections, and verification slips.
 Tenant Records Master directory of stallholders with lease terms, rent collection, and stallkeeper rosters.	 Applicant Management Full application pipeline from intake through document verification to stall assignment.
 Utility Billing Electricity and water meter-based billing engine with receipt printing and batch processing.	 Violations Citation register with demerit point tracking, resolution workflow, and repeat offender monitoring.
 Analytics Formal executive operations report for the Municipal Mayor and Market Board.	 AI Assistant Offline intelligent querying assistant for quick answers about rent, stalls, and billing.

1.3 System Requirements

REQUIREMENT	SPECIFICATION
Operating System	Windows 10 or later (64-bit)
Storage	Local browser storage (data saved to <code>%APPDATA%\Tanauan Public Market System</code>)
Internet	Not required — fully offline operation
Printer	Any A4-capable printer for receipts, reports, and verification slips
Display	Minimum 1280×720 resolution recommended



CHAPTER 2

System Interface & Navigation

2.1 Application Layout

The application uses a modern two-column layout consisting of:

- **Left Sidebar** — Fixed navigation panel (240px wide) with the deep royal blue-to-violet gradient background. Contains your main navigation modules and utility actions.
- **Main Content Area** — The right panel that displays the currently selected module's content, with a sticky top bar at the top.

2.2 Sidebar Navigation

The sidebar is divided into three sections:

Municipal Branding (Top)

Displays the official Municipal Seal of Tanauan, Leyte and the application name "**Tanauan Public Market**" with the subtitle "**Market Office**".

Primary Navigation Modules (Center)

Ten main navigation items are available. The currently active module is highlighted with a warm gold pill background and bold white text:

MODULE	ICON	PURPOSE
Dashboard		Central command center with market overview
Stall Management		Registry of all physical market stalls
Tenant Records		Stallholder directory and rent ledger
Applicants		Application pipeline management
Utility Billing		Electricity & water billing engine
Violations		Compliance and citation register
Analytics		Executive operations report
Logbook		Operational audit trail
Officers		Organizational board and officer directory

MODULE	ICON	PURPOSE
Assistant		Offline AI querying assistant

Utility Actions (Bottom)

At the bottom of the sidebar, you will find:

-  **Settings** — System configuration, backup, and data management
-  **Support** — Help center, FAQ, and contact information
-  **Log Out** — Opens a confirmation dialog before logging out

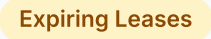
2.3 Top Bar & Search

The sticky top bar (64px height) contains the following elements from left to right:

- **Context Search Bar** — A search input that adapts its placeholder text to the current module (e.g., "Search stalls by ID, tenant, section..." in Stall Management). In the Dashboard, typing here triggers a **global multi-entity search** across Tenants, Stalls, Applicants, Bills, Violations, and Officers.
- **Quick Action Button** — On certain modules, a context-specific action button appears (e.g., " " on the Dashboard).
- **Philippine Standard Time Clock** — A live digital clock displaying the current time in  format along with the full date, updating every second.
- **Notification Bell** — Displays a red dot when there are active alerts. Click to open the notifications panel.
- **Admin Avatar** — Circular avatar displaying "AD" for the system administrator.

2.4 Notifications Panel

Clicking the notification bell  opens a dropdown panel that displays real-time operational alerts. These notifications are automatically generated by the system based on current data:

ALERT TYPE	TRIGGER CONDITION
	Tenants whose rent is past due and uncollected
	Electricity or water bills that are unpaid or overdue
	Applications awaiting review
	Tenant leases approaching expiration

ALERT TYPE	TRIGGER CONDITION
Maintenance	Stalls currently under maintenance
Open Violations	Unresolved violation citations

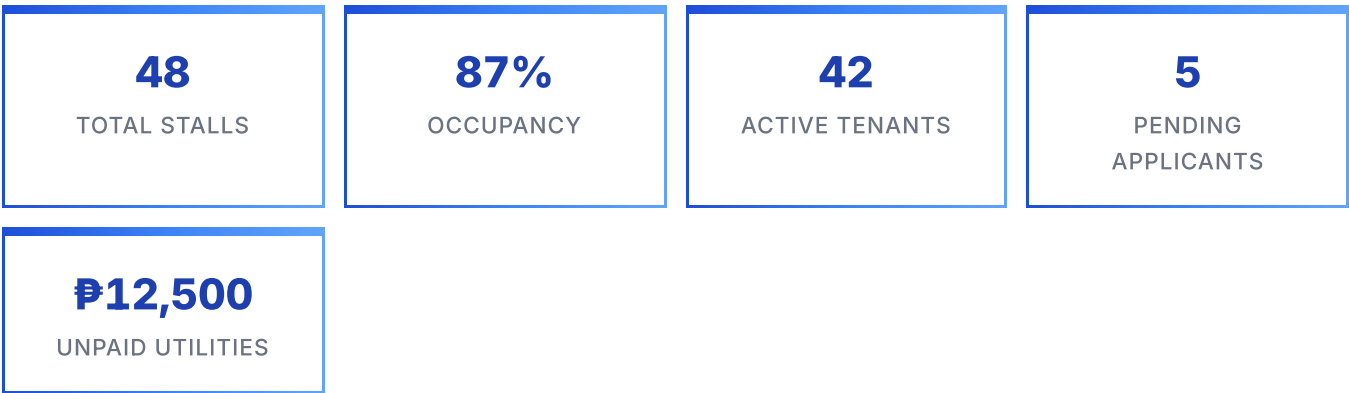
CHAPTER 3

Dashboard

The Dashboard is the central operational command center of the system. It provides a real-time, high-level overview of all market operations at a glance. This is the first page you see after logging in.

3.1 Statistics Cards

The top row displays five key metric cards, each showing a primary figure with supporting breakdowns:



CARD	PRIMARY VALUE	BREAKDOWN DETAILS
Total Stalls	Count of all registered stalls	Occupied, Available, Temp. Closed, Closed, Abandoned, Maintenance
Occupancy	Occupied count / Total stalls	Progress bar showing occupancy percentage
Active Tenants	Total active tenant records	Number expiring soon; quick link to "View Tenants"
Pending Applicants	Awaiting review count	Incomplete documents count; quick link "Review Now"
Unpaid Utilities	Total peso outstanding	Count of unpaid and overdue bills; link to Utilities

3.2 Market Earnings Panel

This panel provides a comprehensive view of the current month's financial standing:

- **Monthly Rent Roll** — Total contracted rent from all active tenants
- **Rent Outstanding** — Uncollected rent for the current period

- **Early-Payment Discounts Given** — Total value of 20% discounts issued
- **Rent Collected to Date** — Actual rent collected for the month
- **Utilities Collected** — Total utility payments received
- **Total Market Earnings** — Combined rent + utilities collected

A large progress meter visualizes the collection percentage. If there are overdue bills, the meter turns **red** to alert officers.

Click the [Open Rent Register](#) button to navigate directly to the Tenant Records module for detailed rent management.

| 3.3 Stall Occupancy Chart

An interactive stacked bar chart displays stall utilization broken down by **market section** (Meat & Poultry, Fish & Seafood, Vegetables & Fruits, Dry Goods). Each bar is color-segmented by stall status with an accompanying legend. Click [Full Report](#) to navigate to the Analytics module.

| 3.4 Recent Activity Feed

A live feed displays the last 50 system operations in reverse chronological order. Each entry shows:

- A colored action icon indicating the type of activity
- A text description of the action (e.g., "Rent collected from Maria Santos for Stall MS-01")
- A relative timestamp (e.g., "Just now", "5 minutes ago")

| 3.5 Global Search

When you type into the search bar from the Dashboard, the system performs a **global multi-entity search**. Results are categorized and displayed in a list of up to 24 items across:

- Tenants · Stalls · Applicants · Utility Bills · Violations · Officers

Clicking on any search result will navigate directly to the relevant record.



CHAPTER 4

Stall Management

The Stall Management module provides a complete registry of all physical market stalls across all sections. It allows you to track stall status, occupancy, business permits, and inspections.

4.1 Viewing & Filtering Stalls

The module opens with summary statistics at the top: **Total Stalls**, **Occupied**, **Available**, **Maintenance**, and **Temporarily Closed**.

Filter Controls

FILTER	OPTIONS
Market Section	All Sections, Meat & Poultry, Fish & Seafood, Vegetables & Fruits, Dry Goods
Status	All Statuses, Occupied, Available, Maintenance, Temporarily Closed, Closed, Abandoned
Search	Search by Stall ID, occupant tenant name, or section

Header Action Buttons

- [Clear Filters](#) — Reset all filters to show all stalls
- [Print List](#) — Prints the currently filtered stall table as a formal register sheet
- [Stall / Space Verification](#) — Opens the verification slip issuance dialog (see Chapter 13)
- [+ New Stall](#) — Opens the Add Stall form

Data Table Columns

COLUMN	DESCRIPTION
Stall ID	Unique identifier (e.g., MS-01, CS-01, FV-01, KK-01)
Section	Market section the stall belongs to
Tenant Name	Name of the current occupant, or "Vacant"
Status	Color-coded status badge (green=Available, blue=Occupied, amber=Maintenance, red=Closed, gray=Abandoned)

COLUMN	DESCRIPTION
Business Permit	Permit badge: With BP No BP Not Recorded
Last Inspection	Date of the most recent stall inspection
Actions	View Details, Issue Verification Slip, Delete

4.2 Adding a New Stall

Click the + New Stall button to open the Add Stall modal form.

1 Enter Stall Details

Fill in the following fields:

- **Stall ID** — Enter a custom ID or leave empty for auto-generation
- **Section** — Select the market section (Meat & Poultry, Fish & Seafood, Vegetables & Fruits, or Dry Goods)
- **Tenant Name** — Required if the status is set to "Occupied"
- **Status** — Select the initial stall status
- **Business Permit** — Select: With BP, No BP, or Not Recorded

2 Save the Record

Click Save to create the stall. The system will automatically assign a unique ID if one was not provided, and a logbook entry will be created.

4.3 Viewing Stall Details

Click the  (eye) icon on any stall row to open the **Stall Details View**. This formal record sheet displays:

- **Stall Particulars** — ID, section, status, permit status, last inspection date
- **Assigned Tenant** — Name, contact, and barangay of the current occupant
- **Stallkeeper Roster** — List of all registered stallkeepers with their relationship to the tenant
- **Utility Summary** — Electricity and water meter numbers and billing overview
- **Billing History** — Table of all utility bills associated with this stall

From this view you can issue a **Verification Slip** or click [Edit Details](#) to modify the stall record.

4.4 Editing a Stall

The Edit Stall form uses a **"blank keeps on-record"** pattern: all inputs appear blank with the current value shown as a placeholder (e.g., *"On record: Meat & Poultry"*). Only fields that you explicitly fill in will be updated.

 **Linked Stall Protection:** When a stall is linked to a tenant record, the tenant name and certain status fields are locked in the stall edit form. Changes to these must be made through the Tenant Records module instead.

Available fields for editing:

- **Section** — Change the market section
- **Status** — Update the stall status
- **Business Permit** — Update permit status
- **Last Inspection** — Set a new date, or use the [Inspected Today](#) shortcut button

 **Releasing a Stall:** If you change the status to "Available" on an occupied stall, the system will ask for confirmation and then remove the occupant tenant record, releasing the stall.

4.5 Deleting a Stall

Click the  (trash) icon to delete a stall. A confirmation dialog will appear.

 **Occupied Stall Protection:** You cannot delete a stall that is currently occupied by an active tenant. You must first release or reassign the tenant through Tenant Records before the stall can be deleted.

CHAPTER 5 Tenant Records

The Tenant Records module is the master directory of all market stallholders. It manages lease terms, contact details, stallkeeper rosters, utility meters, and the monthly rent collection ledger.

5.1 Managing Tenants

Top Statistics

Five summary cards are displayed at the top:

- **Total Tenants** — Count of all tenant records
- **Active Tenants** — Currently active lease holders
- **Expiring Soon** — Tenants with leases approaching expiration
- **Monthly Revenue** — Total contracted rent roll with average rent per stall
- **Rent Collected** — Collection total for the active period with a progress bar

Filter Controls

FILTER	OPTIONS
Section	All Sections, or specific market section
Status	All Statuses, Active, Expiring Soon
Rent Status	All Rent, Rent Paid, Rent Unpaid, Rent Overdue
Rent Month	Month-year picker for selecting the rent period to view/collect
Search	Tenant name, stall ID, tenant ID, or stallkeeper name

Adding a New Tenant

Click [+ New Tenant](#) to open the Add Tenant form with the following fields:

FIELD	DESCRIPTION	VALIDATION
Name	Full name of the tenant	Required
Mobile Number	Contact phone number	11-digit validation
Barangay	Tenant's barangay of residence	—

FIELD	DESCRIPTION	VALIDATION
Stall ID	Autocomplete dropdown from available stalls	Required; must be available
Section	Auto-filled based on selected stall	—
Monthly Rent	Contracted monthly rent amount in pesos	Required; numeric
Status	Active or Expiring Soon	—
Rent Due Day	Day of month when rent is due (1–28)	Default: 5
Electricity Meter No.	Electricity meter number for this stall	—
Water Meter No.	Water meter number for this stall	—
Stallkeepers	Embedded stallkeeper editor (see Section 5.4)	—

5.2 Monthly Rent Collection

The Tenant Records table includes an interactive rent collection feature. For the selected **Rent Month** (set via the month picker filter), each row displays:

- A **checkbox** — Tick to record the tenant's rent as paid for that month
- A **status badge** — Shows Paid, Unpaid, or Overdue
- A **discount tag** — Shows the early-payment discount if applicable

1 Select the Rent Month

Use the **Rent Month picker** in the filter bar to choose which month you are collecting rent for. You can select past months for retroactive logging or future months for advance payments.

2 Mark Rent as Paid

Click the **checkbox** next to the tenant's name. The system will immediately record the payment with today's date and calculate any applicable early-payment discount.

3 Verify the Payment

The status badge will change to Paid. If the payment qualifies for an early discount, you will see a -₱X early tag.

 **Undo Payment:** If a payment was recorded in error, simply uncheck the checkbox to reverse the recorded payment for that month.

5.3 Early-Payment Discount (20%)

The system automatically applies a **20% early-payment discount** when rent is paid on or before the tenant's due date:

Discount Calculation Rule

- Each tenant has a configurable **Rent Due Day** (1 to 28, default: 5th of the month)
- If rent is collected **on or before** the due day of the month → **20% discount applied**
- The net collected amount = $\text{Monthly Rent} \times 0.80$
- Both the *net collected amount* and the *discount amount* are stored permanently
- If the due day has not yet passed, the table shows an incentive tag: **[P] if paid today**

5.4 Stallkeeper Management

Each tenant can have multiple stallkeepers — individuals authorized to operate the stall on behalf of the tenant. Stallkeepers are managed through an embedded editor within the Add/Edit Tenant forms.

Stallkeeper Fields

FIELD	DESCRIPTION
Name	Full name of the stallkeeper
Relationship	Self, Spouse, Child, Parent, Sibling, Other Relative, or Hired Helper
Contact Number	Phone number
Barangay	Barangay of residence

Viewing Tenant Details

Click the  icon on any tenant row to open the **Tenant Details View**, a formal record sheet that includes:

- **Tenant Particulars** — Name, ID, phone, barangay, assigned stall
- **Stallkeeper Information** — All registered stallkeepers and their details

- **Utility Meters** — Electricity and water meter numbers
- **Rent Account** — Payment date, discount received, and days late (if overdue)
- **Rent Ledger** — Up to 12 months of rent payment history
- **Bill History** — All utility bills associated with this tenant

CHAPTER 6 Applicant Management

The Applicant Management module tracks the entire application pipeline — from intake and document verification through approval or rejection, and seamless transition to stall assignment.

6.1 Application Pipeline

Top Statistics

- **Total Applicants** — All applications on record
- **Pending Review** — Complete applications awaiting officer review
- **Incomplete Docs** — Applications with missing documentary requirements
- **No Business Permit** — Compliance flag for applicants without a business permit
- **Approved (This Month)** — Applications approved in the current month

Application Status Flow



i Automatic Status Advancement: The application status automatically progresses as requirements are ticked. When all 6 documentary requirements are submitted, the status changes from "Incomplete" to "Pending Review" automatically.

6.2 Documentary Requirements

Each applicant must submit **6 documentary requirements**. The system tracks submission status with a visual checklist:

#	REQUIREMENT	DESCRIPTION
1	Business Permit	Valid business permit from BPLO
2	Barangay Clearance	Clearance from the applicant's barangay
3	Community Tax Certificate	Cedula (CTC) for the current year
4	Health / Sanitary Permit	Health certificate or sanitary permit
5	2×2 ID Photo	Recent identification photograph

#	REQUIREMENT	DESCRIPTION
6	Signed Contract of Lease	Executed lease agreement with the Market Office

In the applicant table, the **Requirements** column shows the completion status as colored dots (e.g., **4/6**) with tooltips listing each specific requirement.



Business Permit Sync: When an applicant who has already been assigned a stall submits their Business Permit, the linked stall's permit status automatically updates from **No BP** to **With BP**.

6.3 Approval & Stall Assignment

When reviewing an applicant, officers can **Approve** or **Reject** the application:

1 Open the Applicant Review Form

Click on the applicant's row or the review action button to open the review form. Here you can edit the applicant's information and tick/untick documentary requirements.

2 Click "Approve"

After verifying all requirements, click the **Approve** button. This immediately launches the **Assign Stall Form**.

3 Assign a Stall

The stall assignment form shows all genuinely vacant stalls in the market. If the applicant stated a preferred stall and it is still available, it will be pre-selected. Fill in:

- **Stall** — Select from available stalls
- **Monthly Rent** — Set the contracted rent amount
- **Section** — Auto-filled based on selected stall

The applicant's name and phone number are automatically carried over to create a new tenant record.

4 Confirm Assignment

Click **Assign & Create Tenant** to finalize. The system will:

- Create a new Tenant record with the applicant's details
- Mark the selected stall as "Occupied"

- Link the applicant record to the new tenant
- Log the action in the Logbook

i Skip Stall Assignment: You can click **Skip for now** to approve the applicant without immediately assigning a stall. The stall can be assigned later.

CHAPTER 7 Utility Billing

The Utility Billing module is the meter reading calculation and billing engine for **Electricity** and **Water** utilities across all market stalls.

7.1 Bill Calculator

The embedded Bill Calculator at the top of the page allows you to create new utility bills.

Utility Rates

UTILITY	RATE	UNIT	LATE SURCHARGE
 Electricity	₱11.50	per kWh	3.35%
 Water	₱25.00	per cu.m	10.00%

Billing Formula

Consumption = Current Reading – Previous Reading (minimum 0)

Amount Due = Consumption × Rate per Unit

How to Create a New Bill

1 Select Utility Type

Toggle between **Electricity** and **Water**. The system automatically loads the correct rate, unit labels, and icon.

2 Select Stall & Tenant

Choose a stall from the dropdown — the occupant tenant, last recorded meter reading, and meter number are auto-filled. Alternatively, select a tenant to auto-select their stall.

3 Set Period Covered

Enter the **Period From** and **Period To** dates. Use the **Cover all of [Month]** button to quickly set the first and last day of the current month.

4 Set Due Date

The due date defaults to 15 days after the period end date. You can adjust this manually.

5 Enter Meter Readings

The **Previous Reading** is auto-populated from the last bill on record. Enter the **Current Reading**. The system validates that readings are not negative and that the current reading is not less than the previous reading.

6 Review the Live Calculation

A real-time summary card displays: stall, tenant, meter number, period range, readings, consumption, rate, and total amount due. If the due date has already passed, a warning shows the computed late surcharge.

7 Save or Print

- **Save to Records** — Posts the bill, writes a Logbook entry, and updates meter info if needed
- **Print Receipt** — Opens the Print Preview for immediate receipt printing
- **Clear** — Resets the calculator

 **Duplicate Detection:** If a bill already exists for the same stall, utility type, and billing period, the system will show a confirmation modal asking whether you want to proceed or cancel.

7.2 Billing Records & Management

Below the calculator, the **Billing Records Table** lists all issued bills with the following filters:

- **Utility Type** — Filter by Electricity or Water
- **Status** — Filter by Unpaid, Paid, or Overdue
- **Search** — Search by bill ID, stall number, tenant name, or meter number

Table Columns

COLUMN	DESCRIPTION
Select	Checkbox for batch printing selection
Bill ID	Unique bill identifier (e.g., UTL-001)
Utility	Electricity or Water
Stall No.	Associated stall ID
Meter No.	Meter number used for reading
Tenant	Name of the billed tenant
Period	Billing period covered
Consumption	Calculated consumption (kWh or cu.m)
Amount	Billed amount in pesos
Status	Paid Unpaid Overdue
Actions	View, Mark Paid/Unpaid, Print Receipt, Delete

7.3 Receipt Printing & Batch Printing

Single Bill Printing

Click the  icon on any bill row to open the **Print Preview Dialog**. Options include:

- **Number of Copies:** 1 copy, 2 copies (Tenant's Copy + Market Office Copy), or 4 copies (adding Treasurer's Copy + File Copy)
- **Zoom Controls:** Slider from 35% to 200%, "Fit" button, and "Actual Size (100%)" button
- **Fullscreen Mode:** For detailed preview (press **Esc** to exit)

Batch Printing

1 Select Bills for Printing

Use the checkboxes on each row to select individual bills, or use the header checkbox to select all visible bills.

2 Review the Batch Bar

A floating batch bar appears at the bottom showing: *"X bills selected · Y A4 sheets, 4 receipts to a sheet"*

3 Print Selected

Click **Print X Receipts**. The system tiles up to 4 different stall receipts per A4 sheet in a 2×2 grid with dotted cut guidelines.

Receipt Layout

Each printed receipt includes: the Municipal Seal, "Republic of the Philippines" header, stall and tenant details, meter readings, consumption calculation, amount due, due date, late surcharge (if applicable), and the "Printed By" officer name.

7.4 Late Surcharges

Late Surcharge Rules

- **Electricity:** 3.35% surcharge on the billed amount after the due date
- **Water:** 10.00% surcharge on the billed amount after the due date
- Surcharges are calculated **dynamically at the time of printing** — they appear on the receipt if the due date has passed, but the original ledger amount remains unchanged
- This ensures historical records are accurate while printed receipts reflect the current total owed

 **"Printed By" Memory:** The system remembers the issuing officer's name from the last print session, so you only need to enter it once. The name is stored in **localStorage** and persists across sessions.



CHAPTER 8

Violations Register

The Violations Register is a compliance and citation tracking module that records tenant offences, assigns demerit points, and manages dispute resolutions.

8.1 Overview & Statistics

- **Total Violations** — All recorded citations
- **Open Citations** — Unresolved violations requiring attention
- **Resolved Citations** — Successfully resolved cases
- **Open Demerit Points** — Total unresolved demerit points across all tenants

Repeat Offenders Panel

A dedicated panel ranks tenants by their total **unresolved demerit points**, helping market inspectors identify repeat offenders and prioritize enforcement actions.

8.2 Standard Citation Issues

The system includes 8 pre-defined violation categories:

#	VIOLATION ISSUE
1	Late document submission
2	Improper stall cleanup
3	Health code violation
4	Unpaid rent or utilities
5	Operating beyond stall boundary
6	Unauthorized subleasing
7	Fire safety non-compliance
8	Obstruction of walkway

8.3 Recording a Violation

1 Click "+ Record Violation"

Opens the violation recording form.

2 Fill in Citation Details

- **Party Cited** — Select from a datalist of tenant names, or type a custom name for non-tenants
- **Demerit Points** — Assign the severity (numeric points)
- **Violation Issue** — Select from the 8 standard issues or enter a custom violation
- **Status** — Open or Resolved
- **Date Recorded** — The date of the violation
- **Date Resolved** — If already resolved, enter the resolution date
- **Officer Notes** — Free-text field for recording details and context

3 Save the Citation

Click [Save](#) to record. A logbook entry is automatically created.

| 8.4 Resolving & Managing Violations

Open the **Edit Violation** form to:

- Change the status from [Open](#) to [Resolved](#) (automatically stamps the resolution date)
- Reopen a resolved citation (clears the resolution date)
- Amend demerit points or officer notes

Additional Features

- [Export CSV](#) — Download all violation records as a CSV file
- [Print List](#) — Print the filtered violations table
- Table is sorted with open cases first, then by newest recorded date



CHAPTER 9

Analytics & Operations Report

The Analytics module generates a formal **executive operations report** intended for the Municipal Mayor and Market Board. The report is compiled live from all current system records and is organized into seven numbered sections.

Report Header

The report masthead includes the Tanauan Municipal Seal, document reference number ([PMRMS-AR-YYYYMMDD](#)), and generation timestamp in Philippine Standard Time.

9.1 Report Sections

SECTION	TITLE	CONTENTS
I	Executive Summary	Headline cards with progress meters: Stall Occupancy Rate, Active Tenancies, Applications Pending, Collection Efficiency Rate, Open Violations & Demerit Points
II	Stall Utilization	Stacked bar chart of stalls by section and status + status summary table with exact counts and proportion bars
III	Applicant Processing	Pipeline table by status (Pending Review, Incomplete, No BP, Approved, Rejected) with counts and filing shares
IV	Tenancy & Rental Coverage	Table by market section: total stalls, occupied, tenant count, monthly rent roll, section occupancy %
V	Utility Billing & Collection	Stacked bar chart + collection standing table (Electricity/Water consumption, billed, collected, outstanding)
VI	Compliance & Violations	Summary metrics + table of recent violations with status badges
VII	Logbook Activity	Breakdown of operational logs by type and share of total logbook entries

9.2 Exporting the Report

Click the [Export Report](#) button to download a comprehensive JSON file ([civic-market-core-report.json](#)) containing all data underlying the report. This can be used for external analysis, archival, or sharing with other offices.

i Live Data: The report is generated from live data. Every time you open the Analytics module, the report reflects the most current state of all records in the system.

CHAPTER 10 Logbook (Audit Trail)

The Logbook serves as the operational audit trail and daily activity ledger. It records inspections, incidents, collections, maintenance activities, and announcements.

| 10.1 Log Types

TYPE	DESCRIPTION	COLOR
Inspection	Stall inspections, permit checks	Blue
Incident	Security incidents, complaints, disturbances	Red
Maintenance	Repairs, cleaning, facility upgrades	Amber
Collection	Rent and utility bill collections	Green
Announcement	Public notices, schedule changes, advisories	Gray

| 10.2 Automatic Logging

-  **System-Generated Entries:** The system automatically writes a structured logbook entry for every CRUD action — adding stalls, collecting rent, issuing utility bills, recording violations, appointing officers, issuing verification slips, and more. This creates a complete, tamper-evident audit trail of all market operations.

| 10.3 Manual Log Entries

- 1 Click "+ New Entry"**
Opens the new log entry form.
- 2 Fill In Details**
 - **Log Type** — Select from Inspection, Incident, Maintenance, Collection, or Announcement
 - **Details** — Enter a description of the activity or event

3 Save

The entry is timestamped with the current date and time and added to the logbook.

| 10.4 Filtering & Export

- **Date Filter** — View logs from a specific recorded date
- **Type Filter** — Filter by log type (Inspection, Incident, etc.)
- **Search** — Search log details and type
- [Export Log](#) — Download all log entries as a CSV file
- [Print List](#) — Print the currently filtered log entries

 **Audit Trail Integrity:** When deleting a log entry, a safety warning reminds you that the logbook serves as an official audit trail. Deletions should only be done when absolutely necessary.



CHAPTER 11

Market Assistant (AI)

The Market Assistant is an offline intelligent querying tool that allows officers to ask plain-language questions about rent, stalls, bills, tenants, and violations — and receive accurate, data-driven answers.

11.1 How It Works

The assistant uses a **two-tier engine** to ensure accuracy:

Tier 1 — Deterministic Pattern Matching

Your question is first analyzed against keyword patterns for 12 core market intents. If a strong match is found, the system pulls the answer directly from the database using precise calculations — *no AI guessing*.

Tier 2 — Local AI Model (Optional)

If keywords don't match clearly, the system may use a local AI model running entirely on your computer to classify your intent. The AI can also rephrase factual answers into natural conversational language.



Data Accuracy Guarantee: All figures (amounts, counts, names) are *never* generated by AI. They are always computed by the system's deterministic business logic. If the AI rephrases an answer and any number or name doesn't match the computed facts, the AI phrasing is automatically discarded and the verified record text is shown instead.

11.2 What You Can Ask

INTENT	EXAMPLE QUESTIONS
Overdue Rent	"Who has overdue rent?", "List overdue tenants"
Unpaid Rent	"Who hasn't paid rent this month?"
Collected Rent	"How much rent has been collected?"
Tenant Payment	"Has Maria Santos paid her rent?"
Total Earnings	"What are total market earnings?"

INTENT	EXAMPLE QUESTIONS
Vacant Stalls	"Which stalls are available?"
Stall Status	"What's the stall occupancy breakdown?"
Unpaid Utilities	"Are there unpaid electricity bills?"
Meter Lookup	"What's the meter number for Stall MS-01?"
Tenant Lookup	"Show me details for Juan Dela Cruz"
Pending Applicants	"How many pending applications?"
Open Violations	"Any open violations?"

11.3 Using the Assistant

- **Suggestion Chips** — Quick-tap pre-written questions for common queries
- **Chat Input** — Type your question in natural language in the composer bar
- **Source Badges** — Each answer shows its source:
 - **From the records** — Direct database query
 - **Worded by model · figures verified** — AI phrased with verified data
 - **Model wording discarded** — AI response rejected; showing raw data
- **Navigation Buttons** — Answers include buttons like "Open Tenants" or "Open Billing" for direct navigation to the relevant module
- **Model Status** — Header badge shows whether a local AI model is active (🧠) or records-only mode (📄)



CHAPTER 12

Officers & Organizational Board

The Officers module manages the organizational hierarchy and directory of municipal market officials, inspectors, collectors, and supervisors.

12.1 Top Statistics

- **Seats on the Board** — Total officer positions defined
- **Officers in Post** — Actively filled positions
- **On Leave** — Officers currently on leave
- **Vacant Seats** — Unfilled positions
- **Offices Represented** — Count of unique offices in the hierarchy

12.2 Dual View Modes

Organizational Board View

An interactive organizational chart tree displayed as a hierarchy with branching connector lines. Each node shows:

- Officer avatar with color gradient
- Name and position title
- Office name
- Status tag: Active, On Leave, or Vacant
- Quick edit button

Directory View

A tabular listing with columns: Officer (Name & Position), Office, Reports To, Contact, Date Appointed, Status, and Actions. Includes an Office filter dropdown and pagination.

12.3 Pre-seeded Board Positions

POSITION	OFFICE
Municipal Mayor	Office of the Municipal Mayor
Market Board Chairperson	Market Board

POSITION	OFFICE
Market Supervisor	Market Office
Assistant Market Supervisor	Market Office
Market Collector	Market Office
Market Inspector	Market Office
Records Officer	Market Office
Sanitary Inspector	Municipal Health Office
Licensing Officer	Business Permit & Licensing Office
Market Treasurer / Cashier	Municipal Treasurer's Office

12.4 Adding & Editing Officers

Click [+ Add Officer](#) to open the officer form with the following fields:

FIELD	DESCRIPTION
Position	Job title (e.g., Market Inspector)
Office	Select from existing offices or enter a custom office name
Held By	Name of the person filling the position
Status	Active, On Leave, or Vacant
Reports To	Select superior officer from an indented tree dropdown
Date Appointed	Date of appointment to the position
Mobile Number	Contact phone number
Email	Email address

i Hierarchy Safety: The system automatically prevents circular dependencies, self-references, and orphaned seats. When an officer is deleted, all their subordinates are automatically reassigned to the deleted officer's superior to maintain a valid organizational tree.



CHAPTER 13

Stall / Space Verification Slips

Verification Slips are official clearance documents issued by the Market Office to stallholders. These slips are presented to the **Business Permit & Licensing Office (BPLO)** for business permit issuance or renewal.

13.1 Accessing the Verification Slip Feature

There are two ways to issue a verification slip:

- From **Stall Management** — Click the **Stall / Space Verification** button in the header, or the  icon on a specific stall row
- From **Stall Details View** — Click the "Issue Verification Slip" button

13.2 Issuing a Verification Slip

1 Select Purpose

Choose either:

- **Renewal** — For existing lessees renewing their business permits (auto-ticks relevant checklist items)
- **Issuance** — For new lessees obtaining their first business permit

2 Fill In Slip Details

- **Issued By** — Market Supervisor name (remembered across sessions)
- **Issued To** — Stallholder's name (auto-filled if launched from a specific stall)
- **Section** — Market section (auto-filled)
- **Stall No.** — Stall number (auto-filled)

3 Check Requirements

Tick the applicable items from the 8-point checklist:

1. Latest Business Permit (Original)
2. DTI Registration
3. OR of Last Monthly Rental Paid
4. OR of Last Electric/Water Bill Paid

5. Meat Inspector's Clearance
6. Existing Lessee
7. Inspection of Stall
8. New Lessee

An "Others" field allows entering additional custom requirements.

4 Select Slips Per Sheet

Choose how many slips to print per A4 sheet: **1**, **2**, or **4** slips.

5 Print

Click [Print](#) to generate the verification slip(s). Each slip includes a unique **control number** (e.g., SSV-001, SSV-002), date issued, and a validity period of **7 days**.

i Control Number Tracking: Every verification slip receives a sequential control number that is stored permanently in the system. Control numbers are never reused, even if slips are deleted, ensuring complete audit traceability.

Printed Slip Contents

The printed verification slip includes:

- Official Tanauan Market Office seal
- Control number
- Purpose (Issuance or Renewal)
- Stallholder name, section, and stall number
- Checked requirements list
- Date Issued and Valid Until dates
- Validity notice: "This slip is valid for seven (7) days from the date of issue"
- Signature line for the Market Supervisor



CHAPTER 14

Settings & Data Management

The Settings module provides system information, database metrics, and tools for backup, import, and reset operations.

14.1 System Information

INFORMATION	DESCRIPTION
Application Version	v1.0
Storage Type	Local Browser Storage
Total Record Count	Aggregate count of records across all data tables
Last Saved	Timestamp of the most recent automatic save

14.2 Quick Stats

A summary card showing current counts: Total Stalls, Active Tenants, Pending Applicants, Utility Bills, Open Violations, and Unpaid Utilities.

14.3 Data Management

Export All Data

Click [Export All Data](#) to download the entire database as a JSON file ([civic-market-core-report.json](#)). This file can be used for external analysis, archival, or transferring data to another workstation.

Import Data

- 1 Click "Import Data"**
A file picker will open.

- 2 Select a Backup File**
Choose a previously exported JSON backup file. The system will validate the file schema.

3 Review the Import Preview

A confirmation modal displays the exact record counts that will be replaced in each data table.

4 Confirm Import

Click [Confirm](#) to replace the current data with the imported records.

⚠ Data Replacement Warning: Importing data will completely replace all current records. Make sure to export a backup of your current data before importing.

Reset to Defaults

Click [Reset to Defaults](#) to purge all data and reload the default seed data. A confirmation dialog with a danger warning will appear. This action:

- Clears all localStorage data ([pmrms-state-v9](#) , [pmrms-id-counters](#))
- Reloads the default market data (pre-seeded stalls, officers, etc.)
- Cannot be undone unless you have a prior backup

CHAPTER 15 Support & Help Center

The Support module provides operator documentation, system verification tools, full backup/restore capabilities, and contact information for technical support.

15.1 Overview Cards

- **User Guide** — Quick reference for common operations
- **Contact Support** — Direct links to the development team
- **System Updates** — Current version information (v1.0)
- **Data Privacy** — Confirmation that the system is 100% local with zero external telemetry

15.2 Backup & Restore

Download Full Backup

Click [Download Full Backup](#) to generate a complete database backup file named [pmrms-backup-YYYY-MM-DD.json](#). This includes all stalls, tenants, applicants, bills, violations, officers, logbook entries, and verification slip records.

 **Regular Backups:** We recommend downloading a backup at least weekly, or before any major data changes. Store backups on a USB drive or separate folder for safekeeping.

Upload Backup File

Click [Upload Backup File](#) to restore the database from a previously downloaded backup. A status alert will confirm successful restoration.

15.3 Contact Information

The development team can be reached through the following contacts:

NAME	EMAIL	PHONE
Jon Jon Albao	jonjonalbao65@gmail.com	+63 919 845 1397
Kyra Joyce Tondo	kyrajocet@gmail.com	+63 916 425 8038
Shila Mae Marteja	shilamaemarteja18@gmail.com	+63 985 452 8964

15.4 Frequently Asked Questions

The Support page includes an accordion-style FAQ section with 11 detailed operational guides covering:

#	FAQ TOPIC
1	How to add a new stall to the market
2	How to review an applicant and check requirements
3	How electricity meter computation works
4	How water meter computation works
5	How late surcharges are applied
6	How to record and resolve a violation citation
7	How to export reports and analytics data
8	How to transfer data between workstations
9	How to issue a verification slip
10	How to manage the organizational board
11	How to back up and restore system data

APPENDIX A

Business Rules Quick Reference

RULE	DESCRIPTION
Monotonic ID Counters	All IDs (STL, TEN, APP, UTL, VIO, OFC, SSV, LOG) are sequentially assigned and never reused — even after deletion. This prevents historical misattribution.
20% Early Rent Discount	Paying rent on or before the due day (default: 5th, max: 28th) earns a 20% discount. The net collected amount and discount value are stored separately as permanent records.
Dynamic Late Surcharges	Electricity: 3.35% surcharge; Water: 10% surcharge. Calculated at print time when the due date has passed. The original ledger amount is never altered.
Stall-Tenant Linking	Assigning a tenant marks a stall as "Occupied". Moving a tenant releases their previous stall to "Available". Deleting an occupied stall is blocked.
Organizational Hierarchy Safety	Circular dependencies, self-references, and orphaned seats are automatically prevented. Deleting an officer reassigns their subordinates to the deleted officer's superior.
Edit Pattern ("Blank Means Keep")	All edit forms open with blank inputs displaying current values as placeholders. Only explicitly typed fields are updated on save.
Verification Slip Validity	All verification slips are strictly valid for 7 days from the date of issuance.
Duplicate Bill Detection	The system warns if a utility bill already exists for the same stall, utility type, and billing period.
Philippine Standard Time	All dates and times use PHT (Asia/Manila, GMT+8). The live clock and all timestamped entries reflect this timezone.
Offline Operation	The system makes zero network requests. All fonts, icons, and data are stored locally. Internet is never required.



APPENDIX B

Troubleshooting & FAQ

PROBLEM	SOLUTION
The application won't start	Ensure Windows 10 or later is installed. Try right-clicking the application shortcut and selecting "Run as administrator". If the issue persists, reinstall the application.
Data appears missing after reopening	Data is stored in <code>%APPDATA%\Tanauan Public Market System</code> . If you cleared your browser/app data, you may need to restore from a backup file using the Support page.
Cannot delete a stall	Stalls that are currently occupied by an active tenant cannot be deleted. First release or reassign the tenant through the Tenant Records module, then delete the stall.
Print output is misaligned	Ensure your printer is set to A4 paper size with Portrait orientation. In the print dialog, set margins to "Default" or "None" and make sure "Scale to fit" is disabled.
Rent discount was not applied	The 20% discount only applies when rent is collected on or before the tenant's due day. Check the tenant's Rent Due Day setting in their record details.
Search returns no results	Ensure your search term matches the field you're searching (name, ID, section, etc.). Try clearing filters first, as active filters may limit search scope.
Transfer data to another computer	Use Support → Download Full Backup to export your data. Copy the backup file to the new computer via USB drive. Open the app on the new computer and use Support → Upload Backup File to restore.
Market Assistant shows "Records only" mode	This means no local AI model is loaded. The assistant still works fully using direct database queries. To enable AI phrasing, place a compatible GGUF model in the <code>resources/models</code> folder.
Notifications bell shows wrong count	Notification counts are calculated live from current data. If counts seem incorrect, check that the underlying records (overdue rent, pending applications, etc.) are up to date.
Application is running slowly	Close other applications to free system resources. If the database has grown very large, consider exporting data for archival and resetting to defaults for the new period.



APPENDIX C

Contact Information

For technical support, bug reports, or feature requests, please contact the development team:

 **Jon Jon Albao**

 jonjonalbao65@gmail.com

 +63 919 845 1397

 **Kyra Joyce Tondo**

 kyrajocet@gmail.com

 +63 916 425 8038

 **Shila Mae Marteja**

 shilamaemarteja18@gmail.com

 +63 985 452 8964



Public Market Rental Monitoring System

Municipality of Tanauan, Leyte · Market Office

Version 1.0 · © 2026 All Rights Reserved